Test Patch

1. Executive Summary

Total Floor Area	120.00 m²
Stories / Height	2 Stories / 6.0 m
RCC Concrete Volume	3.24 m³
Reinforcement Steel	505 kg
Clay Modular Bricks	6,264 Nos
Masonry & Plaster Cement	56 Bags (50kg)
Masonry & Plaster Sand	13.16 Tonnes
Plaster Surface Area	217.8 m²
Total Estimated Cost	INR 501,923.52

2. Design Basis

Design Codes & Standards:
- IS 456:2000 (Code of Practice for Plain and Reinforced Concrete)

Material Properties:
- Concrete: M25 (Density: 25 kN/m3, E = 5000sqrt(fck))
- Steel: Fe415 (Density: 78.5 kN/m3, E = 2x10^5 MPa)

Loading & Analysis:
- Design Load: User Input (Factored)
- Load Path: Tributary Area Method (Corner: 25%, Edge: 50%, Interior: 100% of bay)
- Beam Analysis: Simplified Simply Supported/Continuous assumptions.
(M = wL^2/8 or wL^2/10, Deflection checked against L/250)

Column Design:
- Short Column Assumption (Le/r < 12)
- Sizing Criteria: Axial Stress Check (Pu/Ag <= 0.4 fck)
Note: Steel capacity (0.67 fy Asc) is reserved as safety margin.

Foundation Design:
- Isolated Square Footings
- Area Required = Load * 1.1 / SBC (1.1 factor for self-weight)
- Depth Check: Punching Shear at d/2 distance (Limit: 0.25sqrt(fck))

Estimation Basis:
- Concrete Cost: INR 5000/m3
- Steel Cost: INR 60/kg
- Rebar Ratios: Columns 150kg/m3, Beams 120kg/m3, Footings 80kg/m3

3. Column Schedule & Quantities (IS 456 Cl. 26.5.3)

ID	Level	Loc (x,y)	Load (kN)	Size (mm)	Conc (m3)	Steel (kg)	Status
C0_L0	L0	(0.00, 0.00)	0.0	300x300	0.27	40.5	PASS
C4_L0	L0	(0.00, 4.00)	0.0	300x300	0.27	40.5	PASS
C8_L0	L0	(0.00, 8.00)	0.0	300x300	0.27	40.5	PASS
C1_L0	L0	(3.30, 0.00)	0.0	300x300	0.27	40.5	PASS
C5_L0	L0	(3.30, 4.00)	0.0	300x300	0.27	40.5	PASS

ID	Level	Loc (x,y)	Load (kN)	Size (mm)	Conc (m3)	Steel (kg)	Status
C9_L0	L0	(3.30, 8.00)	0.0	300x300	0.27	40.5	PASS
C2_L0	L0	(6.60, 0.00)	0.0	300x300	0.27	40.5	PASS
C6_L0	L0	(6.60, 4.00)	0.0	300x300	0.27	40.5	PASS
C10_L0	L0	(6.60, 8.00)	0.0	300x300	0.27	40.5	PASS
C3_L0	L0	(9.90, 0.00)	0.0	300x300	0.27	40.5	PASS
C7_L0	L0	(9.90, 4.00)	0.0	300x300	0.27	40.5	PASS
C11_L0	L0	(9.90, 8.00)	0.0	300x300	0.27	40.5	PASS

4. Reinforcement Schedule (IS 456 Cl. 26.5)

Col ID	Main Bars	Links (Stirrups)
N/A	Run Detailing	First

5. Foundation Schedule & Quantities (IS 456 Cl. 34)

Col ID	Load (kN)	Size (m)	Depth (mm)	Conc (m3)	Steel (kg)
N/A	-	-	-	-	-

6. Beam Schedule & Quantities (IS 456 Cl. 26.5.1)

ID	Size (mm)	Span (m)	Conc (m3)	Steel (kg)	Top Steel	Bot Steel	Stirrups
N/A	-	-	-	-	-	-	-

7. Slab Schedule & Quantities

ID	Thk (mm)	Area (m2)	Conc (m3)	Steel (kg)	Main Mesh	Dist Mesh
N/A	-	-	-	-	-	-

8. Brickwork Schedule & Material Takeoff (IS 1077, IS 2250)

Modular clay bricks 200×200×100 mm with mortar (190×190×90 mm solid core, 250 bricks/m³ per IS 1077). Mortar 1:4 mix proportion with 1.33 dry volume factor per IS 2250. W:C ratio = 0.50.

Wall ID	Fl	Span / Cols	Len (m)	Ht (m)	Thk (mm)	Gross Vol (m³)	Bricks (Nos)	Brick Vol (m³)	Wet Mortar (m³)	Dry Mortar (m³)	Cem Bags	Cem (kg)	Sand (T)	Water (L)
W1_F0	F0	Bay	10.00	2.48	230	5.693	1,424	4.627	1.066	1.418	9	408.3	1.815	204
W2_F0	F0	Bay	12.00	2.48	230	6.831	1,708	5.549	1.282	1.705	10	490.9	2.182	245
W1_F1	F1	Bay	10.00	2.48	230	5.693	1,424	4.627	1.066	1.418	9	408.3	1.815	204
W2_F1	F1	Bay	12.00	2.48	230	6.831	1,708	5.549	1.282	1.705	10	490.9	2.182	245
TOTAL	**2Fl**	**All Walls**	**-**	**-**	**-**	**25.05**	**6,264**	**20.35**	**4.70**	**6.24**	**36**	**1,798**	**7.99**	**899**

9. Plastering Schedule & Quantities (IS 1661, IS 2402, IS 1200)

External wall plastering: 15mm thick in 1:4 cement:sand mix (IS 2402). Internal wall plastering: 12mm thick in 1:6 mix (IS 1661). Ceiling soffit plastering: 6mm thick in 1:3 mix per IS 1661 Cl. 12.3. Dry volume factor = 1.33.

Surface ID	Fl	Location & Mix Spec	Area (m²)	Thk (mm)	Mix	Wet Vol (m³)	Dry Vol (m³)	Cem Bags	Cem (kg)	Sand (T)	Water (L)
W1_EXT_F0	F0	Ext Render (15mm 1:4)	24.75	15	1:4	0.371	0.494	3	142.2	0.632	71

Surface ID	Fl	Location & Mix Spec	Area (m²)	Thk (mm)	Mix	Wet Vol (m³)	Dry Vol (m³)	Cem Bags	Cem (kg)	Sand (T)	Water (L)
W1_INT_F0	F0	Int Wall (12mm 1:6)	24.75	12	1:6	0.297	0.395	2	81.3	0.542	41
W2_EXT_F0	F0	Ext Render (15mm 1:4)	29.70	15	1:4	0.446	0.593	4	170.6	0.758	85
W2_INT_F0	F0	Int Wall (12mm 1:6)	29.70	12	1:6	0.356	0.474	2	97.5	0.650	49
W1_EXT_F1	F1	Ext Render (15mm 1:4)	24.75	15	1:4	0.371	0.494	3	142.2	0.632	71
W1_INT_F1	F1	Int Wall (12mm 1:6)	24.75	12	1:6	0.297	0.395	2	81.3	0.542	41
W2_EXT_F1	F1	Ext Render (15mm 1:4)	29.70	15	1:4	0.446	0.593	4	170.6	0.758	85
W2_INT_F1	F1	Int Wall (12mm 1:6)	29.70	12	1:6	0.356	0.474	2	97.5	0.650	49
Ceil_F0	F0	Slab Soffit (6mm 1:3)	120.00	6	1:3	0.720	0.958	7	344.7	1.149	172
Ceil_F1	F1	Slab Soffit (6mm 1:3)	120.00	6	1:3	0.720	0.958	7	344.7	1.149	172
TOTAL	**2Fl**	**All Plaster Surfaces**	**457.8**	**-**	**-**	**4.380**	**5.826**	**34**	**1,673**	**7.46**	**836**

10. Painting & Surface Finishes Schedule (NBC 2016 SP7)

Surface preparation & coatings per NBC 2016 SP7: Water-thinnable primer (1 coat @ 8 m²/L), polymer wall putty (2 coats @ 1.5 kg/m²), interior acrylic emulsion (2 coats @ 12 m²/L), exterior weather-guard acrylic emulsion (2 coats @ 10 m²/L).

Surface ID	Fl	Location & Specification	Area (m²)	Primer 1-Ct (L)	Putty 2-Ct (kg)	Int Em 2-Ct (L)	Ext Guard 2-Ct (L)
W1_EXT_F0	F0	External Weather Guard	24.75	3.09	—	—	4.95
W1_INT_F0	F0	Int Acrylic Emulsion	24.75	3.09	74.25	4.12	—
W2_EXT_F0	F0	External Weather Guard	29.70	3.71	—	—	5.94
W2_INT_F0	F0	Int Acrylic Emulsion	29.70	3.71	89.10	4.95	—
W1_EXT_F1	F1	External Weather Guard	24.75	3.09	—	—	4.95
W1_INT_F1	F1	Int Acrylic Emulsion	24.75	3.09	74.25	4.12	—
W2_EXT_F1	F1	External Weather Guard	29.70	3.71	—	—	5.94
W2_INT_F1	F1	Int Acrylic Emulsion	29.70	3.71	89.10	4.95	—
Ceil_F0	F0	Slab Soffit (White Emulsion)	120.00	15.00	360.00	20.00	—
Ceil_F1	F1	Slab Soffit (White Emulsion)	120.00	15.00	360.00	20.00	—
TOTAL	**2Fl**	**All Surfaces (Walls + Ceilings)**	**457.8**	**57.2**	**1,046.7**	**58.1**	**21.8**

11. Consolidated Masonry & Finishes BOQ & Cost (IS 1905, CPWD DSR)

11.1 Comprehensive Bill of Materials

Item / Material Description	Specification / Mix Proportion	Building Total	Unit	IS Standard
Total Wall Gross Volume	L × t × Hv (clear storey height)	25.05	m³	IS 1905:1987
Clay Modular Building Bricks	Standard modular 20×20×10 cm	6,264	Nos	IS 1077:1992
Net Solid Brickwork Volume	Actual brick body vol. (excl. mortar)	20.35	m³	IS 1077:1992
Wet Brickwork Mortar Volume	Gross Wall Vol – Solid Brick Vol	4.70	m³	IS 2250:1981
Dry Brickwork Mortar Volume	Wet Volume × 1.33 dry factor	6.24	m³	IS 2250:1981
Brickwork Cement	1:4 Mix Proportion (OPC/PPC)	36 (1,798 kg)	Bags 50kg	IS 269:2015
Brickwork Sand	Coarse sand, dry density 1600 kg/m³	7.99	Tonnes	IS 383 Zone II
Brickwork Water	Water:Cement ratio = 0.50	899	Litres	IS 456 Cl 5.4
Wall Plaster Area	Int (12mm 1:6) + Ext (15mm 1:4)	217.8	m²	IS 1200 Pt 12

Item / Material Description	Specification / Mix Proportion	Building Total	Unit	IS Standard
Ceiling Soffit Plaster Area	Soffit of slabs (6mm 1:3 mix)	240.0	m²	IS 1661:1972
Total Plaster Surface Area	Combined wall + ceiling surface	457.8	m²	IS 1200 Pt 12
Plastering Cement	Combined wall & ceiling plaster	34 (1,673 kg)	Bags 50kg	IS 1661/IS 2402
Plastering Sand	Clean plaster sand (Zone III/IV)	7.46	Tonnes	IS 383:2016
Plastering Water	W:C = 0.50	836	Litres	IS 456 Cl 5.4
Surface Primer	1 coat @ 8 m²/L	57.2	Litres	NBC 2016 SP7
Polymer Wall Putty	2 coats @ 1.5 kg/m²	1,046.7	kg	NBC 2016 SP7
Interior Acrylic Emulsion	2 coats @ 12 m²/L	58.1	Litres	NBC 2016 SP7
Exterior Weather Emulsion	2 coats @ 10 m²/L	21.8	Litres	NBC 2016 SP7
GRAND TOTAL CEMENT (Mortar+Plaster)	**All masonry + plastering combined**	**70 bags (3,471 kg)**	**Bags**	**IS 269:2015**
GRAND TOTAL SAND (Mortar+Plaster)	**All masonry + plastering sand**	**15.46**	**Tonnes**	**IS 383:2016**
GRAND TOTAL WATER (Mortar+Plaster)	**All masonry + plastering water**	**1,736**	**Litres**	**IS 456 Cl 5.4**

11.2 Trade-wise Cost Takeoff

Trade / Material Component	Quantity	Rate (INR)	Subtotal (INR)
Clay Modular Bricks (IS 1077)	6,264 Nos	■9 / brick	■56,376.00
Brickwork Mortar Cement	36 Bags	■380 / bag	■13,680.00
Brickwork Mortar Sand	7.99 Tonnes	■1,500 / Tonne	■11,989.83
Plastering Work (12/15/6mm)	457.8 m²	■280 / m²	■128,184.00
Surface Finishes (Primer+Putty+Paint)	457.8 m²	■120 / m²	■54,936.00
TOTAL MASONRY, PLASTER & FINISHES			**■265,165.83**

12. Structural Layout Plan

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Column Coordinates: C0_L0: (0.00, 0.00) C1_L0: (3.30, 0.00) C2_L0: (6.60, 0.00) C3_L0: (9.90, 0.00)
C4_L0: (0.00, 4.00) C5_L0: (3.30, 4.00) C6_L0: (6.60, 4.00) C7_L0: (9.90, 4.00) C8_L0: (0.00, 8.00)
C9_L0: (3.30, 8.00) C10_L0: (6.60, 8.00) C11_L0: (9.90, 8.00)
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13. Integrated Steel Breakdown

Bar Diameter	Total Weight (kg)
10 mm	0.0
Est_Beam	0.0
Est_Col	505.4
TOTAL	505.4

14. Structural Audit Report

No audit results available.

15. Sustainability Audit

Design Strategy: Standard (OPC)

Material	Quantity	Emission Factor	Total CO2e (kg)
Concrete	3.24 m3	300 kg/m3	972.0

Steel	505 kg	1.85 kg/kg	935.1
TOTAL	-	-	1907.1

Wait! You can reduce your Carbon Footprint.
Refining the design to use **Fly Ash based Concrete (PPC)** instead of OPC can reduce concrete emissions by approximately 15-20%.
Switch "Use Green Concrete" ON in the dashboard to see savings. (Note: PPC is slower setting and has lower early strength, use only if confirmed by RMC supplier).

16. Serviceability Checks (Deflection)

No beam serviceability checks performed.

17. Staircase Design Schedule (Dog-Legged)

No staircase selected for design.

20. Concrete Material Breakdown (IS 10262:2019)

Material	Per m^3	Total
Wet Concrete Volume		3.24 m^3
Cement (50kg bags)	7.6 bags	24.6 bags
Fine Aggregate (Sand)	0.419 m^3	1.36 m^3
Coarse Aggregate (20mm)	0.821 m^3	2.66 m^3
Water	171 kg	554 kg
Water:Cement Ratio	0.45	—
Nominal Mix Ratio	1:1.76:3.13	—

Note: Indicative quantities per IS 10262:2019. Add 3-5% wastage for actual procurement.

24. Professional Disclaimer

IMPORTANT NOTICE - READ CAREFULLY

This structural design report is generated by automated software for **PRELIMINARY DESIGN AND EDUCATIONAL PURPOSES ONLY**.

Key Limitations:
• The software uses simplified analysis methods (tributary area, short column assumptions).
• Seismic forces (IS 1893) and Wind loads (IS 875 Pt 3) have been incorporated for basic checks, but require full dynamic analysis for high-rise buildings.
• Site-specific soil conditions, groundwater, and other geotechnical factors are not evaluated.
• Local building codes may have additional requirements beyond IS 456:2000.

Required Actions Before Construction:
• All designs MUST be reviewed and sealed by a licensed Professional Engineer (PE).
• Independent structural analysis should be performed using appropriate software.
• Geotechnical investigation and foundation recommendations are required.
• All applicable permits and approvals must be obtained.

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Generated by StructOptima. This report is a preliminary design aid compliant with IS 456:2000, IS 1893:2016, IS 13920:2016.
All designs must be verified by a licensed Structural Engineer prior to construction.
Safety Factors: γc = 1.5 (concrete), γs = 1.15 (steel) | Load Factor: 1.5 (DL + LL)

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